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MATT Engineering Solution offers hardware and software courses that prepare students and professionals for today's tech world. We provide hands-on training in embedded systems, microcontrollers, and robotics, along with software skills in programming, data science, web development, and cloud computing.

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DIY QUADCOPTER DRONE

Introduction

- This DIY project focuses on building a quadcopter drone using the components provided in the given document such as a 450 mm frame, 1000 KV BLDC motors, 30A ESC, 2200 mAh battery, flight controller, and transmitter-receiver module.
- A quadcopter drone is one of the most commonly used unmanned aerial vehicles (UAVs) because of its simple design and stable flight performance.
- In this project, the drone is designed and assembled step by step by connecting all hardware components properly and configuring the flight controller.
- The system demonstrates how thrust is generated using propellers and how balance is maintained using motor speed control.
- This DIY implementation helps in understanding real-time drone working and practical engineering concepts such as aerodynamics, embedded systems, and control mechanisms.

Aim and Objectives

- The main aim of this project is to design and develop a working quadcopter drone using the given components.
- The objective is to understand how each component functions and how they are integrated to form a complete drone system.
- This project also focuses on practical learning such as hardware assembly, wiring connections, controller programming, and system testing. Another objective is to ensure safe operation by performing proper calibration and testing procedures.

- By completing this project, users gain hands-on knowledge of drone technology and system integration.

Components Used in DIY Drone

The main components used in this project include a 450 mm quadcopter frame, 1000 KV BLDC motors, 30A Electronic Speed Controllers (ESC), 2200 mAh Li-Po battery, flight controller, propellers, and a 2.4 GHz transmitter and receiver module. The frame acts as the base structure that holds all components. Motors generate thrust required for flying. ESC controls motor speed based on signals from the flight controller. The battery supplies power to all components. The flight controller processes signals and maintains stability. The transmitter and receiver enable wireless control of the drone.

Working Principle of Quadcopter

The quadcopter works based on the principle of lift and thrust generation. When the motors rotate the propellers at high speed, they push air downward, which creates an upward force known as lift. This lift helps the drone to take off from the ground. The flight controller continuously adjusts the speed of each motor to maintain balance and control movement. Two propellers rotate clockwise and two rotate anticlockwise to balance torque. By varying motor speed, the drone can move forward, backward, left, right, and also rotate.

1. QUADCOPTER FRAME DIAGRAM

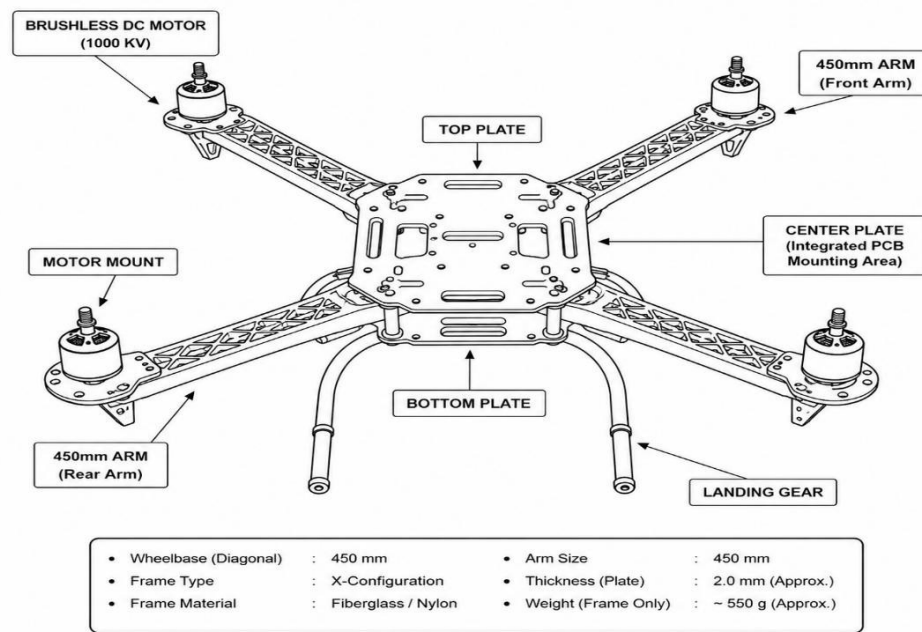


Fig 1: Quadcopter Frame Diagram

Frame Design and Mounting

The 450 mm frame is selected because it provides sufficient space to mount all components such as motors, ESC, battery, and flight controller. The frame must be strong, lightweight, and properly balanced to support stable flight. During assembly, the frame is placed on a flat surface and all arms are fixed tightly using screws. Proper alignment of the frame ensures equal distribution of weight, which is very important for maintaining the center of gravity.

Motor and Propeller Setup

The drone uses four 1000 KV BLDC motors, which are mounted on each arm of the frame. These motors are responsible for generating the thrust required for flight. Each motor is connected to a propeller that pushes air downward to lift the drone. Two propellers rotate clockwise and two rotate anticlockwise. This arrangement helps in maintaining balance and prevents the drone from spinning uncontrollably. Proper installation of propellers is very important, as incorrect placement can cause instability or crash.

2. MOTOR AND PROPELLER DIAGRAM

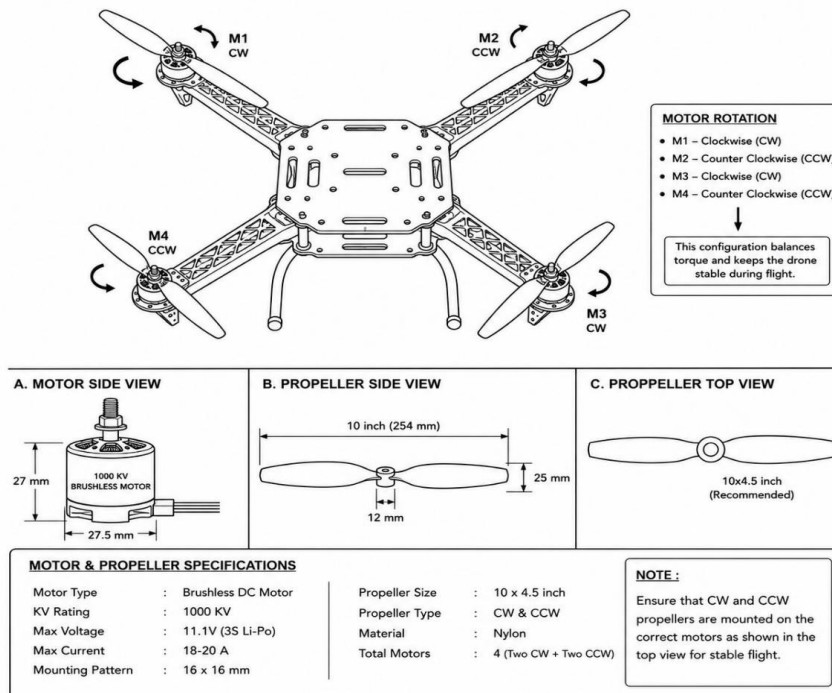


Fig 2: Motor and Propeller Diagram

6. PROPELLER DIRECTION DIAGRAM

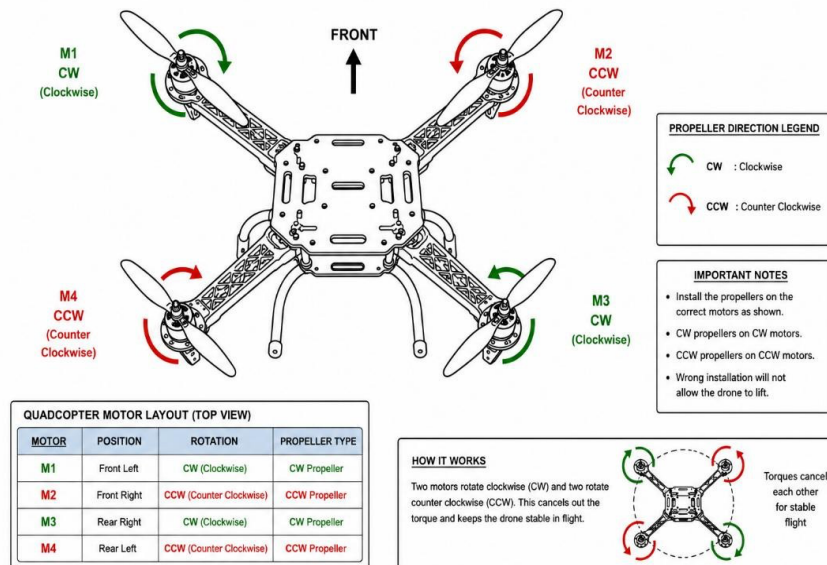
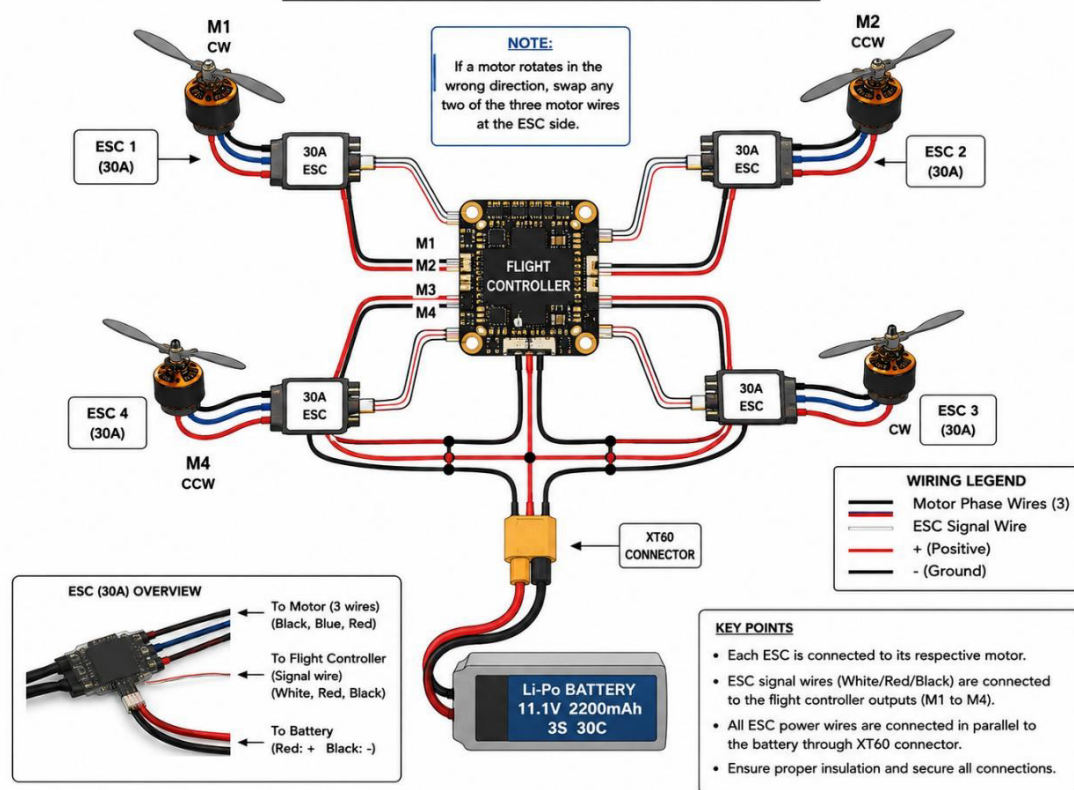


Fig 6: Propeller Direction Diagram

ESC and Wiring Connection

Each motor is connected to a 30A ESC, which controls the speed of the motor based on signals from the flight controller. The ESC receives power from the battery and distributes it to the motors accordingly. All wiring connections must be done carefully to avoid short circuits. The ESC signal wires are connected to the flight controller output pins. Proper wiring ensures smooth communication between components and stable drone operation.

3. ESC CONNECTION DIAGRAM



Flight Controller Installation and Programming

The flight controller is installed at the center of the frame because it acts as the brain of the drone. It receives input from the receiver and sensors and controls motor speed accordingly. After installation, firmware is uploaded to the flight controller. Sensors such as accelerometer and gyroscope are calibrated. Flight

modes and control settings are configured to ensure smooth operation. Proper programming helps the drone respond accurately to user commands.

4. FLIGHT CONTROLLER DIAGRAM

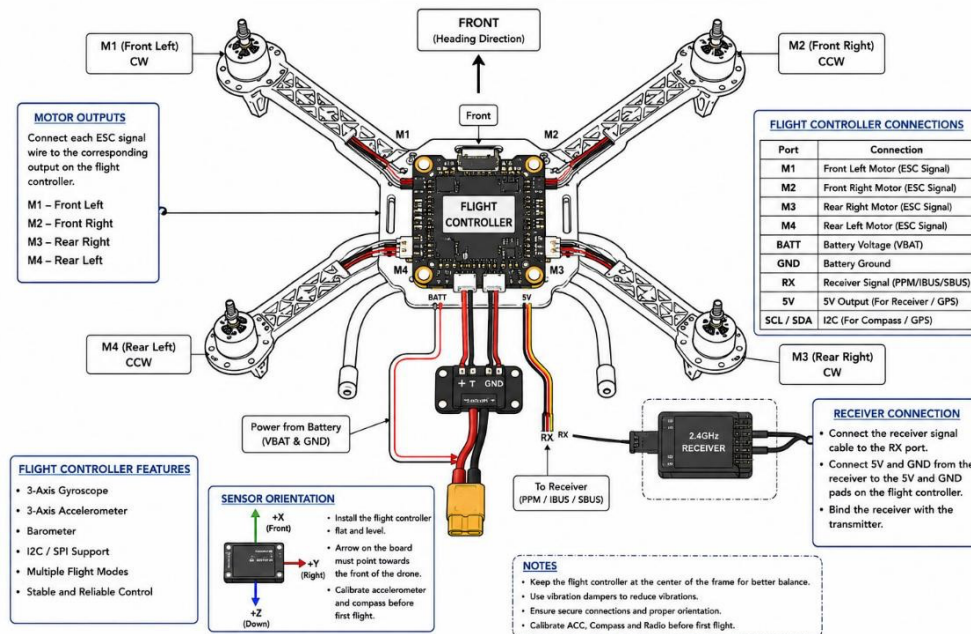


Fig 4: Flight Controller Diagram

Battery and Communication System

A 2200 mAh Li-Po battery is used to supply power to the drone. It is securely fixed on the frame using a strap to avoid movement during flight. The battery capacity determines the flight time of the drone. The 2.4 GHz transmitter and receiver module is used for communication. The transmitter sends control signals, and the receiver receives them and passes them to the flight controller.

5. BATTERY CONNECTION DIAGRAM

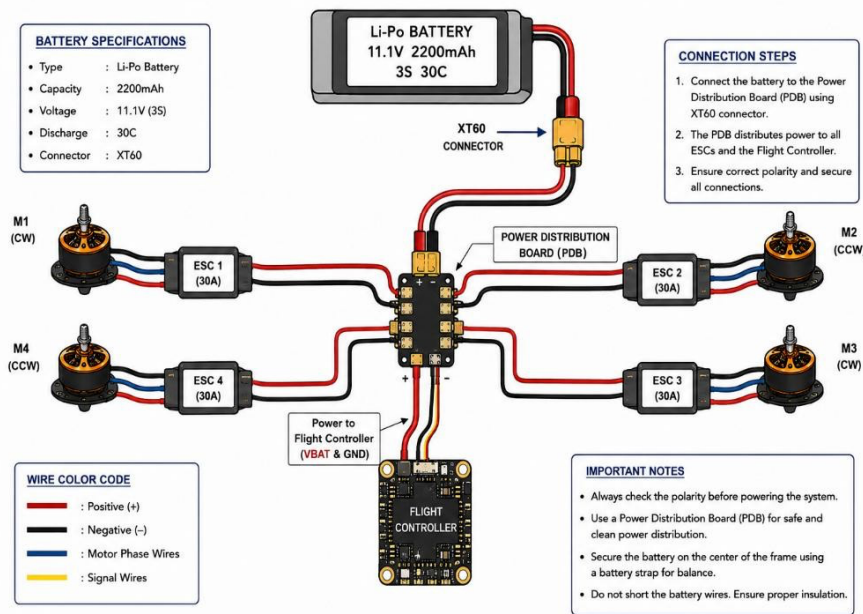


Fig 5: Battery Connection Diagram

Testing and Final Output

After completing assembly and programming, the drone is tested in an open area. All connections are checked before starting the drone. A small lift test is performed first to ensure stability. If the drone shows imbalance or vibration, adjustments are made in tuning. Once stable, full flight testing is performed.

DJI F450 QUADCOPTER HOW TO BUILD?

The DJI F450 quadcopter is a popular and beginner-friendly drone platform used for DIY drone projects. It uses a 450 mm frame with four motors, making it stable, easy to assemble, and suitable for learning drone technology. The build process involves assembling the frame, mounting BLDC motors, connecting ESCs, installing the flight controller, and fixing the battery and propellers. Once all components are properly connected and calibrated, the drone can be controlled using a transmitter for smooth takeoff, flight, and landing. This project helps in understanding the practical working of drone systems and their real-time applications.



DIY QUADCOPTER DRONE USER MANUAL

Before You Start

Before assembling the quadcopter, ensure all components are available and working properly.

Required Components

- F450 Quadcopter Frame
- 4 × BLDC Motors
- 4 × ESC (Electronic Speed Controller)
- KK2.1.5 Flight Controller
- 2200mAh Li-Po Battery
- 2 × Clockwise Propellers

- 2 × Counterclockwise Propellers
- Transmitter and Receiver
- Connecting Wires
- Battery Strap
- Screwdriver
- Soldering Iron

STEP 1 — Frame Assembly

Instructions

1. Place the bottom frame plate on a flat surface.
2. Attach all four arms to the frame using screws.
3. Tighten all screws properly.
4. Ensure the frame is balanced and stable.

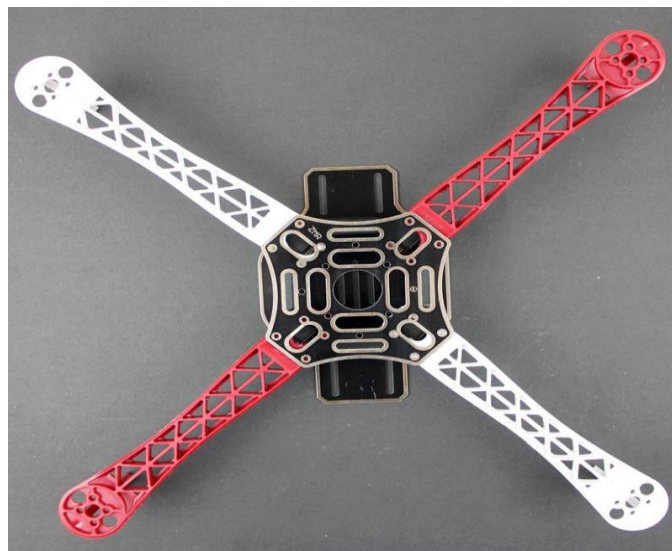


Fig F450 Quadcopter Frame

Important Note

- Do not overtighten screws because it may damage the frame.

STEP 2 — ESC Installation

Instructions

1. Take all four ESCs.
2. Solder the red wire to the positive (+) terminal.
3. Solder the black wire to the negative (-) terminal.
4. Fix one ESC on each arm using zip ties.



Fig ESC (electronic speed controller)

Important Note

- Incorrect polarity connection may permanently damage the ESC.

STEP 3 — Motor Mounting

Instructions

1. Place one BLDC motor on each arm.
2. Fix the motors using screws.
3. Connect motor wires to the ESC wires.
4. Ensure all connections are tight.



Fig BLDC Motors

Important Note

- Loose motor connections may cause unstable flight.

STEP 4 — Flight Controller Installation

Instructions

1. Place the KK2.1.5 flight controller at the center of the frame.
2. Fix it using double-sided tape or Velcro.
3. Ensure the white arrow on the controller points toward the front side of the drone.
4. Connect ESC signal wires to the controller output pins.



Fig Flight Controller Installation

Important Note

- Wrong controller direction will cause incorrect drone movement.

STEP 5 — Receiver Connection

Instructions

1. Mount the receiver on the drone frame.
2. Connect receiver channels to the flight controller.
3. Verify the following channel connections:

Channel	Function
CH1	Roll
CH2	Pitch
CH3	Throttle
CH4	Yaw
CH5	AUX



Fig transmitter



Fig receiver

STEP 6 — Battery Installation

Instructions

1. Place the Li-Po battery at the bottom center of the frame.
2. Secure the battery using the battery strap.
3. Connect the battery connector carefully.



Fig Battery

Warning

- Never puncture or damage the Li-Po battery.

STEP 7 — ESC Calibration

Instructions

1. Turn ON the transmitter.
2. Move the throttle stick to maximum position.
3. Press and hold Button 1 and Button 4 on the KK2.1.5 board.
4. Connect the battery.
5. Wait for beep sounds.
6. Move throttle to minimum position.
7. Wait for confirmation beeps.

Result

- ESC calibration is now completed.

STEP 8 — Sensor Calibration

Instructions

1. Place the drone on a perfectly flat surface.
2. Open the KK2.1.5 menu.
3. Select “ACC Calibration”.
4. Wait until calibration completes.

Important Note

- Do not move the drone during calibration.

STEP 9 — Motor Direction Check

Instructions

1. Arm the drone.
2. Slowly increase throttle.
3. Observe motor rotation direction.

Correct Rotation

Motor Rotation

Motor 1 Clockwise

Motor 2 Counterclockwise

Motor 3 Clockwise

Motor 4 Counterclockwise

If Rotation is Wrong

- Swap any two motor wires connected between motor and ESC.

STEP 10 — Propeller Installation

Instructions

1. Install clockwise propellers on clockwise motors.
2. Install counterclockwise propellers on counterclockwise motors.
3. Tighten propeller nuts properly.



Fig Propeller

Warning

- Incorrect propeller installation may cause drone crash.

STEP 11 — Initial Flight Test

Instructions

1. Move to an open outdoor area.
2. Turn ON the transmitter.
3. Connect the battery.
4. Arm the drone.
5. Slowly increase throttle.
6. Check drone stability before full flight.

Safety Instructions

- Keep away from people and animals.
- Do not fly near electric wires.
- Fly only in open areas.

Drone Control Guide

Control Function

Throttle Up / Down

Pitch Forward / Backward

Roll Left / Right

Yaw Rotation

Maintenance Instructions

- Check propellers before every flight.
- Recharge battery safely.
- Keep electronics dry.
- Tighten loose screws regularly.

Safety Warnings

- Do not touch spinning propellers.
- Use only compatible batteries.
- Avoid flying during rain or strong wind.
- Disconnect battery after use.



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